

A reaction $A + B \rightarrow C$ obeys the following rate equation:

$$\text{Rate} = k[A]^3$$

- (a) If $[B]$ is doubled, how will the rate change? Will the rate constant change?
- (b) What are the reaction orders for A and B? What is the overall reaction order?
- (c) What are the units of the rate constant?

For the reaction $2 \text{N}_2\text{O}_5 \rightarrow 4 \text{NO}_2 + \text{O}_2$. At 55°C , the rate equation is first order for N_2O_5 and the rate constant is $4.12 \times 10^{-3} \text{ s}^{-1}$.

(a) Write the rate law for the reaction.

(b) What is the rate of reaction when $[\text{N}_2\text{O}_5] = 0.050 \text{ M}$?

(c) What happens to the rate when the concentration of N_2O_5 is tripled to 0.150 M ?

(d) What happens to the rate when the concentration of N_2O_5 is reduced by 10 % to 0.045 M ?

Consider the following reaction: $\text{C}_2\text{H}_5\text{Cl} + \text{OH}^- \rightarrow \text{C}_2\text{H}_5\text{OH} + \text{Cl}^-$, the rate law for this reaction is first order in $\text{C}_2\text{H}_5\text{Cl}$ and first order in OH^- . When $[\text{C}_2\text{H}_5\text{Cl}] = 4.0 \times 10^{-3} \text{ M}$ and $[\text{OH}^-] = 2.5 \times 10^{-2} \text{ M}$, the reaction rate at 310 K is $5.20 \times 10^{-2} \text{ M s}^{-1}$.

- (a) What is the value of the rate constant? What are its units?
- (b) What would happen to the rate if the concentration of OH^- were cut in half?
- (c) What would happen to the rate if the concentration of both reactants were cut in half?

The following data were measured for the reaction $\text{BF}_3 + \text{NH}_3 \rightarrow \text{F}_3\text{BNH}_3$

Experiment	$[\text{BF}_3] \text{ (M)}$	$[\text{NH}_3] \text{ (M)}$	Initial Rate (M/s)
1	0.250	0.250	0.2130
2	0.250	0.125	0.1065
3	0.200	0.100	0.0682
4	0.350	0.100	0.1193
5	0.175	0.100	0.0596

- (a) What is the rate law for the reaction?
- (b) What is the overall order of the reaction?
- (c) Calculate the rate constant with units
- (d) What is the rate when $[\text{BF}_3] = 0.100 \text{ M}$ and $[\text{NH}_3] = 0.500 \text{ M}$?

The initial rate of the reaction $A+B \rightarrow C+D$ is determined for different initial conditions, with the results listed in the table.

Experiment	[A] (M)	[B] (M)	Initial rate (Ms^{-1})
1	0.185	0.144	3.35×10^{-4}
2	0.185	0.288	1.35×10^{-3}
3	0.370	0.144	6.75×10^{-4}
4	0.370	0.288	2.70×10^{-3}

- a) What is the order of reaction with respect to A and to B?
- b) What is the overall reaction order?
- c) What is the value of the rate constant, k?

For a reaction $A + B \rightarrow \text{products}$, where $\text{rate} = k [A]^x$

Determine the half life of reactant A, $t_{1/2}$, from the differential rate equations of zeroth, first and second order reactions.

Does the half-life always depend on the initial concentration of the reactant?

The kinetics of a reaction $A + 5B + 6C \rightarrow 3D + 3E$ is being studied. The data in the following table is obtained:

Experiment	Initial [A] (M)	Initial [B] (M)	Initial [C] (M)	Initial rate (M/s)
1	0.35	0.35	0.35	8.0×10^{-4}
2	0.70	0.35	0.35	3.2×10^{-3}
3	0.70	0.70	0.35	6.4×10^{-3}
4	0.70	0.35	0.70	3.2×10^{-3}

- Calculate the partial order for all reactants. For full marks make sure you show all your calculations and explain your reasoning.
- Calculate the overall order of the reaction
- Do you think that this reaction can be a single step reaction?

a) Identify the order of the following reactions (partial and overall orders) assuming that they are single step reactions:



b) Discuss how you can determine these orders experimentally

c) Discuss how you can determine the rate constants experimentally

d) Discuss how you can determine the activation energy of these reactions experimentally

e) How likely is it that a reaction like shown in iii will be a single step reaction?

f) How is it possible that a reaction is of 0 order? Discuss.

A reaction $A + B + C \rightarrow D$ follows the multistep pathway shown below: (F is an intermediate).

Step 1 is an equilibrium, which is established quickly. The rate-determining step of the reaction is step 2.



a) Determine the order for all reactants and the overall order

b) Write a rate equation for this reaction based exclusively on the concentration of reactants.

A reaction $A + B \rightarrow C$ (single step process) can be carried out in the presence of a catalyst D to speed up the rate of the reaction. Propose an equation for the reaction rate of A and B in the presence of D. Discuss the limitations of your calculations.

Calculate the rate constant for the reaction catalysed by C and for the reaction without catalyst if $[A] = 0.185 \text{ M}$, $[B] = 0.185 \text{ M}$, $[C] = 9.25 \times 10^{-3} \text{ M}$ for an initial rate of $3.35 \times 10^{-4} \text{ Ms}^{-1}$

The reaction $A + B \rightarrow C + D$ occurs in two steps:

Step 1. $A \rightarrow F$ Slow

Step 2. $F + B \rightarrow C + D$ Fast

- a) Which one is the rate determining step?**
- b) What is the partial order for every reactant? What is the reaction order overall? Write a rate equation for the reaction.**
- c) The rate constant for the reaction is $2.16 \times 10^{-3} \text{ s}^{-1}$. Calculate the half life of reactant A.**
- d) Draw an energy diagram for the reaction. Identify the reactants, products, intermediates, Transition states and Energies of activation.**

A reaction $A + B \rightarrow C$ (single step, k_1 rate constant) can also be carried out in the presence of a catalyst D. When D is used the reaction mechanism is:

Step 1 (quickly established equilibrium, K equilibrium constant): $A + D \rightleftharpoons E$

Step 2 (irreversible reaction, k_2 rate constant): $E + B \rightarrow C + D$.

- a) Show the equation for the reaction rate for the non-catalysed reaction (without D)
- b) Show the equation for the reaction rate for the catalysed reaction with D (for full marks show carefully all your calculations)
- c) What is the relationship between k_1 and k_2K if D is a catalyst?

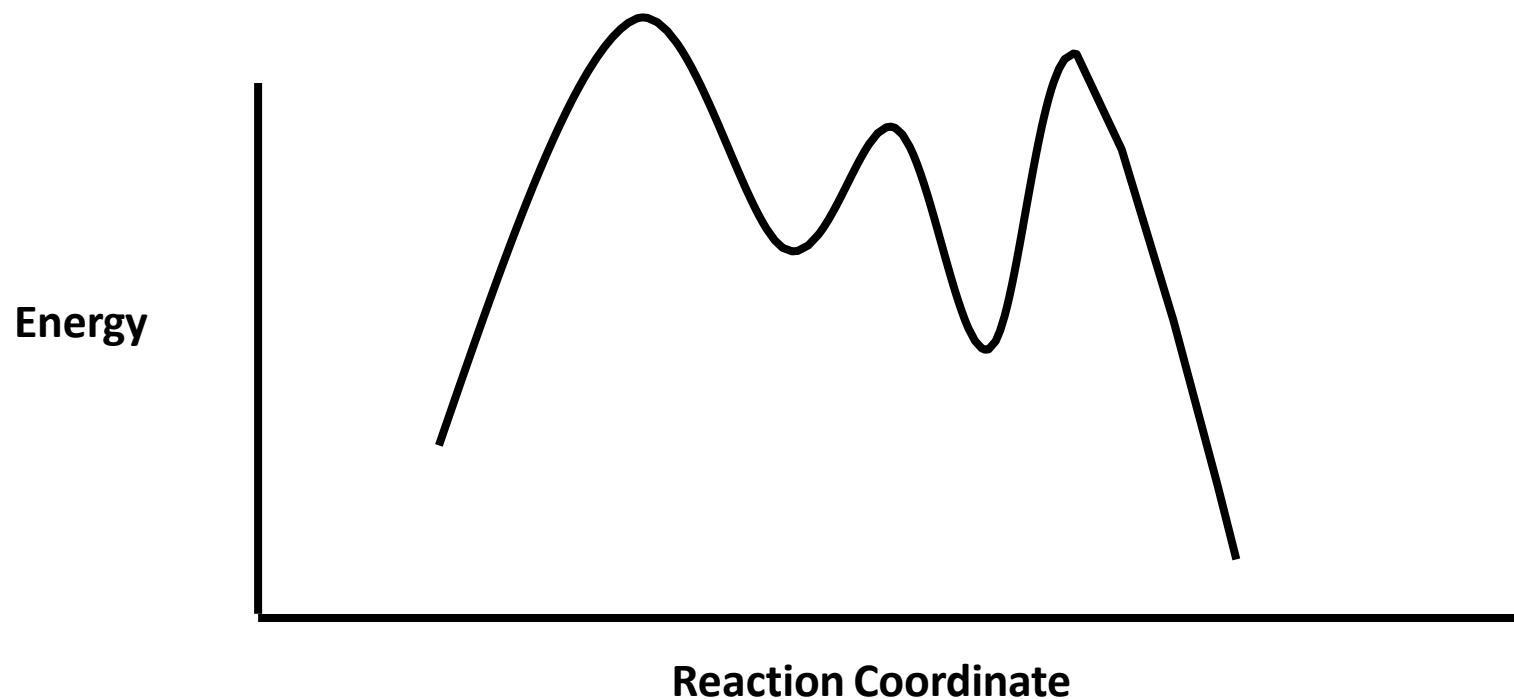
The reaction $A + B \rightarrow C$ occurs in three steps:

Step 1. $A + B \rightarrow F$

Step 2. $F \rightarrow G$

Step 3. $G \rightarrow C$

- a) Place in the diagram below all the reactants, products, intermediates and transition states
- b) How many intermediates and transition states are there in this reaction?
- c) Which step is the rate limiting step and why?
- d) What is the rate law, partial orders for the reactants and overall reaction order?



The data in the following table is obtained for the kinetics of the reaction
 $2A + 3B + C \rightarrow 4D + 2E + F$:

Experiment	Initial [A] (M)	Initial [B] (M)	Initial [C] (M)	Initial Rate (M/s)
1	0.15	0.15	0.15	2.0×10^{-3}
2	0.60	0.15	0.15	2.0×10^{-3}
3	0.15	0.45	0.15	6.0×10^{-3}
4	0.15	0.45	0.30	6.0×10^{-3}

- a) Calculate the partial order for all reactants.
- b) Calculate the overall order of the reaction
- c) Calculate the rate constant
- d) Calculate the half life for reactant B



15 minutes